

Landscape of Machine Learning in Databases

ML Workloads



Regression



Classification



Clustering



Anomaly
Detection



Recommendation

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In -Database ML Capabilities

Feature
Engineering

Model Training
(Linear/GLM)

Automatic
Differentiation

Model
Inference

Gradient-Based
Optimization

Database Engine Layer

SQL Processing

Query Optimizer

Vectorized
Execution
Engine

Storage
Manager

Transaction
Manager

Storage Layer

Tables(Data)

Indexes

Metadata

Statistics

Logs

Benefits



Data
Locality



Reduced
Data
Movement



Privacy
and
Security



Performance



Scalability